

MIST4610

Group Project

Case#1 – The Wilderness Exploration Society

The Wilderness Exploration Society (WES) at Peachtree State University (Georgia) wants to develop a database to manage equipment rentals and trip registrations. The WES director has provided the following information about the current situation:

- Customers of WES include Peachtree State University members (students, faculty/staff, and alumni) as well as non-university customers (also known as guests). To participate in WES activities, a guest must be sponsored by a Peachtree State University member. A member can sponsor many guests, but each guest has only one sponsor. Once a guest has a sponsor, they may rent equipment and participate in trips just like any other customer. Basic information is recorded for all customers and sponsors, including: ID number, name, phone number, and date of birth.
- Customers can take part in a variety of trips such as backpacking, kayaking, mountain biking, rock climbing, or camping. Each trip type has a name and a code (e.g., the “Beginner Kayaking” trip has trip code “4123”). Trip types are also defined by a difficulty level, a fee, and a length (measured in days).
- Some trip types have prerequisites. A trip type may require completion of one or more other trip types (for example, “Intermediate Kayaking” might require “Beginner Kayaking”). A trip type may be a prerequisite for many other trip types. WES needs to record these prerequisite relationships.
- Each trip type may be scheduled on multiple dates during the year. For example, the “Beginner Kayaking” trip might be scheduled for April 20, 2026, May 20, 2026, and June 12, 2026. However, a given trip type cannot be scheduled more than once on the same date.
- All registrations for scheduled trips (both past and present) must be recorded. For each scheduled trip registration, it should also be noted whether or not a participant has signed the required liability waiver. In addition, WES records a participant’s registration status as one of: confirmed, waitlisted, or cancelled. A participant can have at most one registration record per scheduled trip.
- Each scheduled trip is led by one WES staff member, with another acting as assistant leader.
- Equipment available for rental includes tents, backpacks, sleeping bags, and so on. Each equipment type has three daily rental rates: one for students, one for faculty/staff/alumni, and one for guests. For example, an equipment type called “Osprey Aether 70L Pack” may be rented by students for \$2.00 per day, by faculty/staff/alumni for \$3.00 per day, and by guests for \$4.00 per day.
- Every individual piece of equipment has a unique item number and a condition rating (good, fair, or poor). For instance, WES might have three “Osprey Aether 70L Pack” items: each has its own number, two are in good condition, and one is in fair condition. Rental costs are determined by equipment type, not by condition.
- All equipment rentals (past and present) must be recorded. A customer may rent one or several items at the same time. When they do, a rental agreement is created. This agreement includes the rental date, the customer’s details, and the list of items rented. The rental date applies to the entire agreement, but each item can have its own expected return date. The actual return date for each item must also be recorded.
- Each rental agreement is managed by a specific WES staff member.
- Finally, note that customers may rent equipment without going on a trip, and they may also go on a trip without renting equipment. These two activities are managed independently.